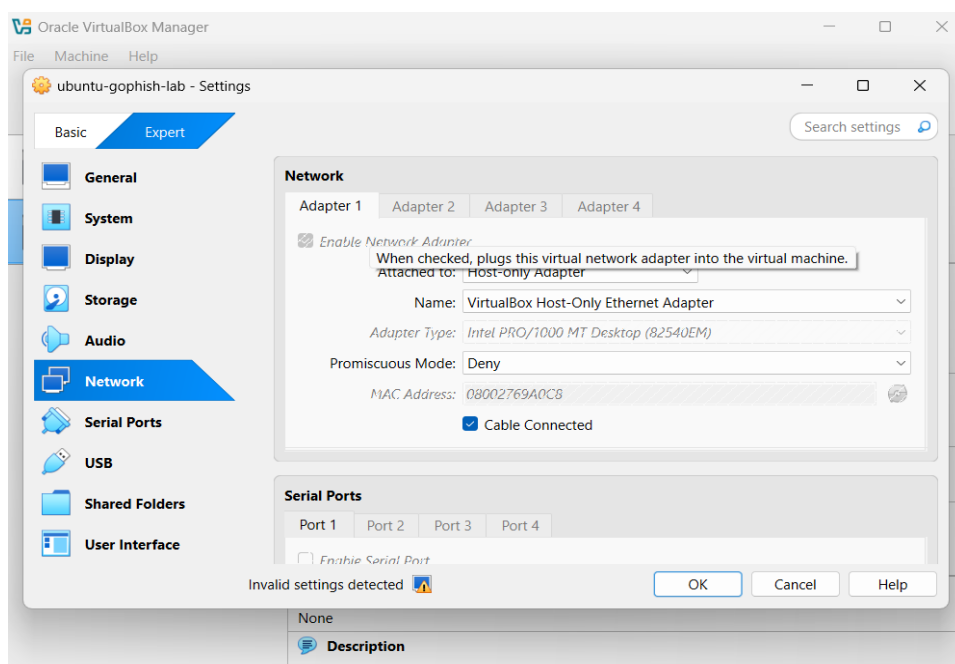
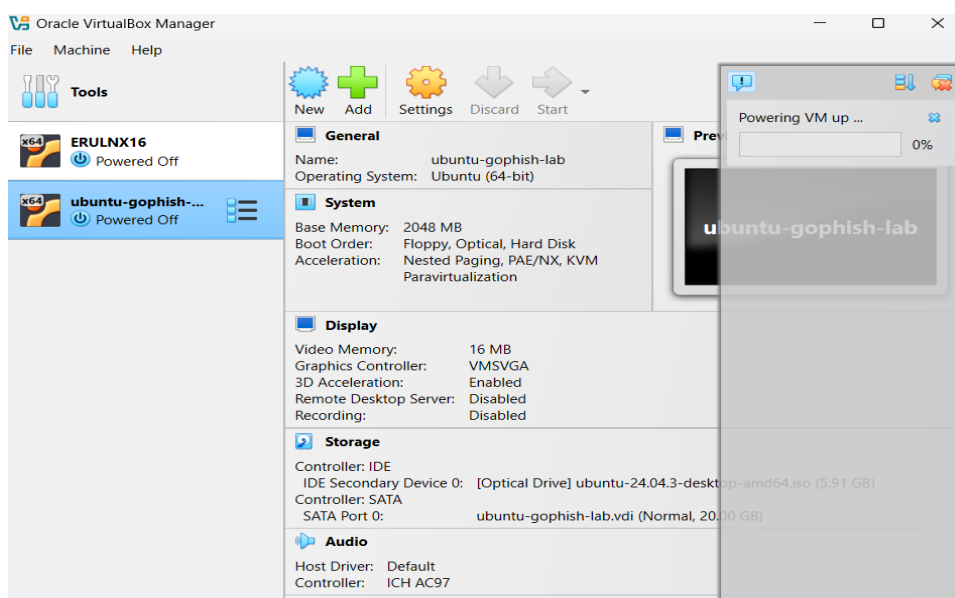


GoPhish Simulation Report

Tools & Environment

- **Host:** Windows
- **Virtualization:** Oracle VirtualBox (VM name: ubuntu-gophish-lab)
- **VM OS:** Ubuntu Desktop (version: [insert])
- **MailHog (local SMTP catcher)** — UI: <http://127.0.0.1:8025>, SMTP: 127.0.0.1:1025
- **Network:** Host-Only / Internal (isolated lab)
- **Snapshots**



1. Executive Summary

This report documents a safe, controlled phishing simulation conducted inside an isolated VirtualBox Ubuntu VM using GoPhish and MailHog. The exercise demonstrates installation, configuration, campaign creation, local delivery capture, and event tracking without contacting external systems or real users. The campaign sent a single test email (captured locally), the link was clicked during testing, and no real credentials were captured.

2. Objective

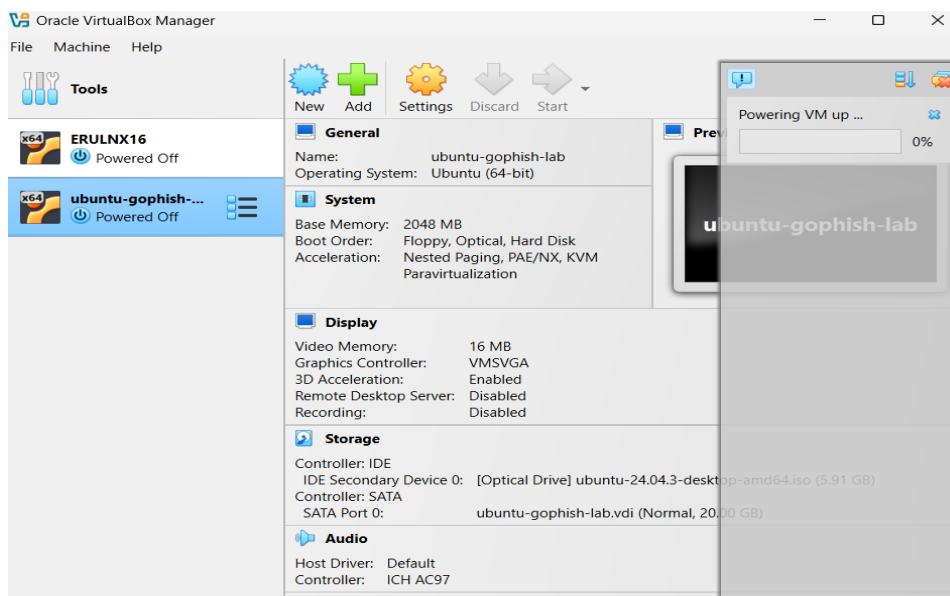
- Demonstrate the safe installation and operation of GoPhish within an isolated VM.
- Create a controlled campaign that sends a training email to a test user and capture the message locally using MailHog.
- Document the workflow and outcomes for educational submission.

3. Safety Controls & Authorization

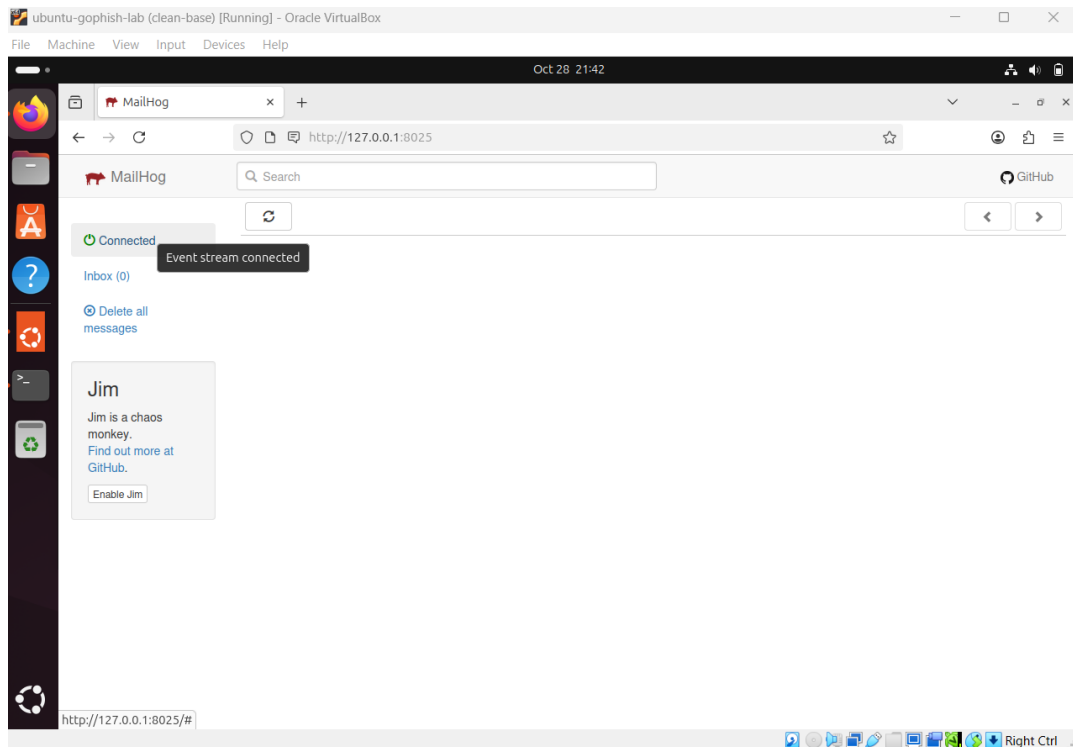
- All testing performed in VM: ubuntu-gophish-lab.
- Network set to Host-Only / Internal to prevent internet egress.
- MailHog used as local SMTP sink — no real email delivery.
- Test accounts used with fake addresses (e.g., testuser@local.test).

4. Methodology (Steps Performed)

1. **VM Preparation:** Created ubuntu-gophish-lab in VirtualBox, installed Ubuntu Desktop, and took snapshot clean-base.



2. **Install Tools:** Ensured basic tools (curl, unzip) present in the VM.
3. **MailHog:** Downloaded and ran MailHog in the VM to capture SMTP locally at 127.0.0.1:1025 and provide a web UI at 127.0.0.1:8025.



4. **GoPhish:** Copied GoPhish to the VM, made the binary executable, and started it. Logged into admin UI at <https://127.0.0.1:3333> and immediately changed the default admin password.

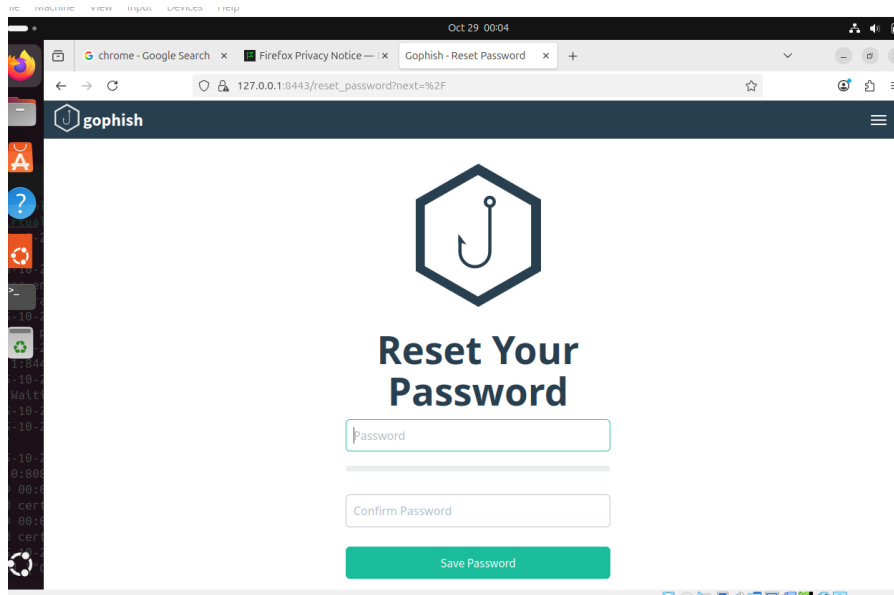
```

ubuntu-gophish-lab (clean-base) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Oct 28 21:42

MailHog
http://127.0.0.1:8025
MailHog
Search
Connected
Event stream connected
Inbox (0)
Delete all messages
Jim
Jim is a chaos monkey.
Find out more at GitHub.
Enable Jim
http://127.0.0.1:8025/#
Right Ctrl

ann@ann-VirtualBox: ~/Downloads
2776 0.0 0.1 17812 2340 pts/0 S+ 23:02 0:00 grep --color=
auto gophish
ann@ann-VirtualBox:~/Downloads$ chmod +x gophish
ann@ann-VirtualBox:~/Downloads$ pkill gophish || true
ann@ann-VirtualBox:~/Downloads$ ./gophish
time="2025-10-28T23:05:28+05:30" level=warning msg="No contact address has been
configured."
time="2025-10-28T23:05:28+05:30" level=warning msg="Please consider adding a con
tact_address entry in your config.json"
goose: no migrations to run. current version: 20220321133237
time="2025-10-28T23:05:28+05:30" level=info msg="Please login with the username
admin and the password 4e08a1f2ea326ad4"
time="2025-10-28T23:05:28+05:30" level=info msg="Starting admin server at https:
//127.0.0.1:3333"
time="2025-10-28T23:05:28+05:30" level=info msg="Background Worker Started Succe
ssfully - Waiting for Campaigns"
time="2025-10-28T23:05:28+05:30" level=info msg="Starting IMAP monitor manager"
time="2025-10-28T23:05:28+05:30" level=info msg="Starting new IMAP monitor for u
ser admin"
time="2025-10-28T23:05:28+05:30" level=info msg="Starting phishing server at htt
p://0.0.0.0:80"
time="2025-10-28T23:05:28+05:30" level=fatal msg="listen tcp 0.0.0.0:80: bind: p
ermission denied"
ann@ann-VirtualBox:~/Downloads$

```



5. **Sending Profile:** Created Sending Profile MailHog-local pointed to 127.0.0.1:1025 (no auth, TLS disabled) and used Send Test Email to verify local delivery.

A screenshot of the 'New Sending Profile' dialog box in Gophish. The dialog has a white background and a dark blue header with the title 'New Sending Profile' and a close button. The form contains the following fields: 'Name' with the value 'Mail-Hog-local'; 'Interface Type' with a dropdown menu showing 'SMTP'; 'SMTP From' with the value 'lab@local.test'; 'Host' with the value '127.0.0.1'; 'Username' with the value 'Username'; and 'Password' with a masked input field showing '*****'. At the bottom, there is a checkbox labeled 'Ignore Certificate Errors' which is checked. The dialog is overlaid on a dark background.

6. **Landing Page:** Created a training landing page (credential capture disabled for safety) in GoPhish.

127.0.0.1:8443/landing_pages

Name:

Training_Landing

Import Site

HTML

```
<html><head></head><body>This is a training page. No credentials collected.</body></html>
```

☐ Capture Submitted Data

Cancel Save Page

- Email Template:** Created an HTML email template using `{{.FirstName}}` and `{{.URL}}` placeholders. Previewed to confirm rendering.

Send Test Email ×

✓ Email Sent!

Send Test Email to:

Test User testuser@local.test Trainee

Cancel Send

- Users & Groups:** Created a test group Test-Employees and added a fake user test@local.test.

Name:

[+ Bulk Import Users](#) [Download CSV Template](#)

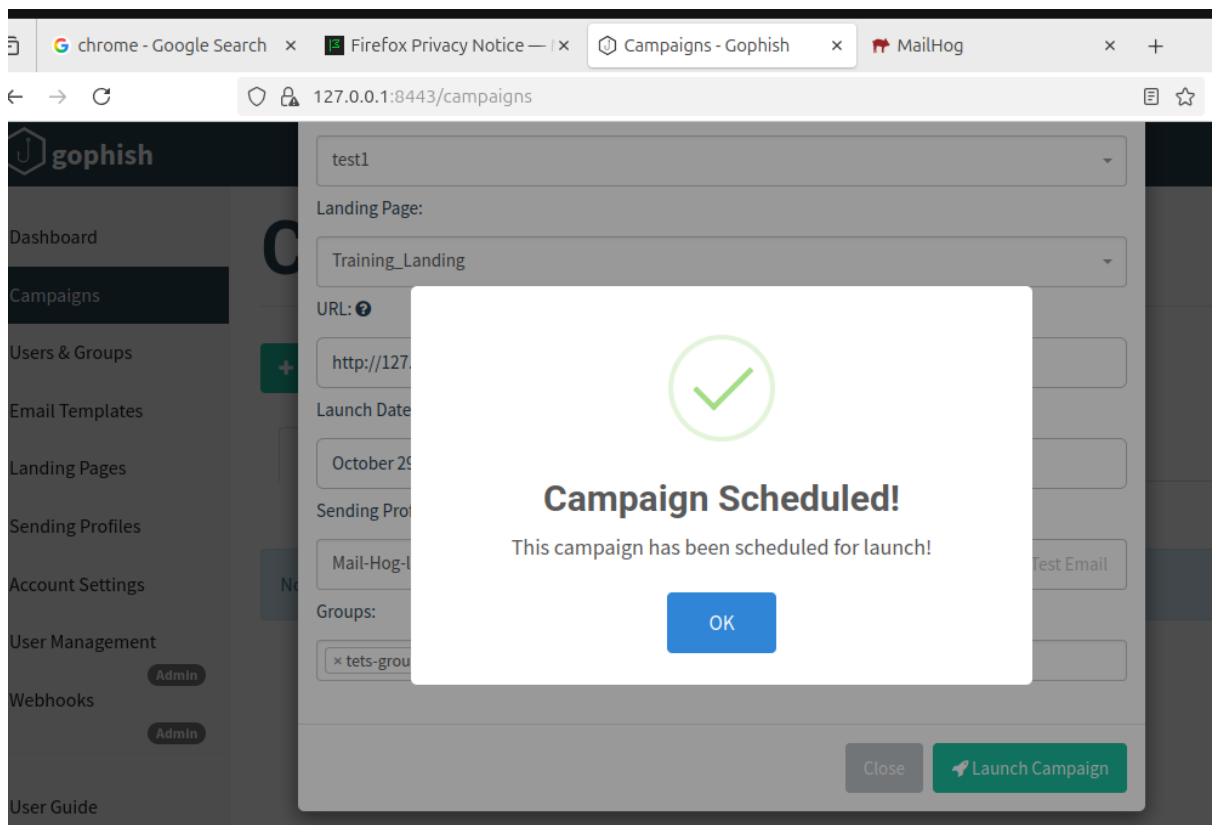
Show entries Search:

First Name	Last Name	Email	Position
Test	User	test@local.test	Trainee

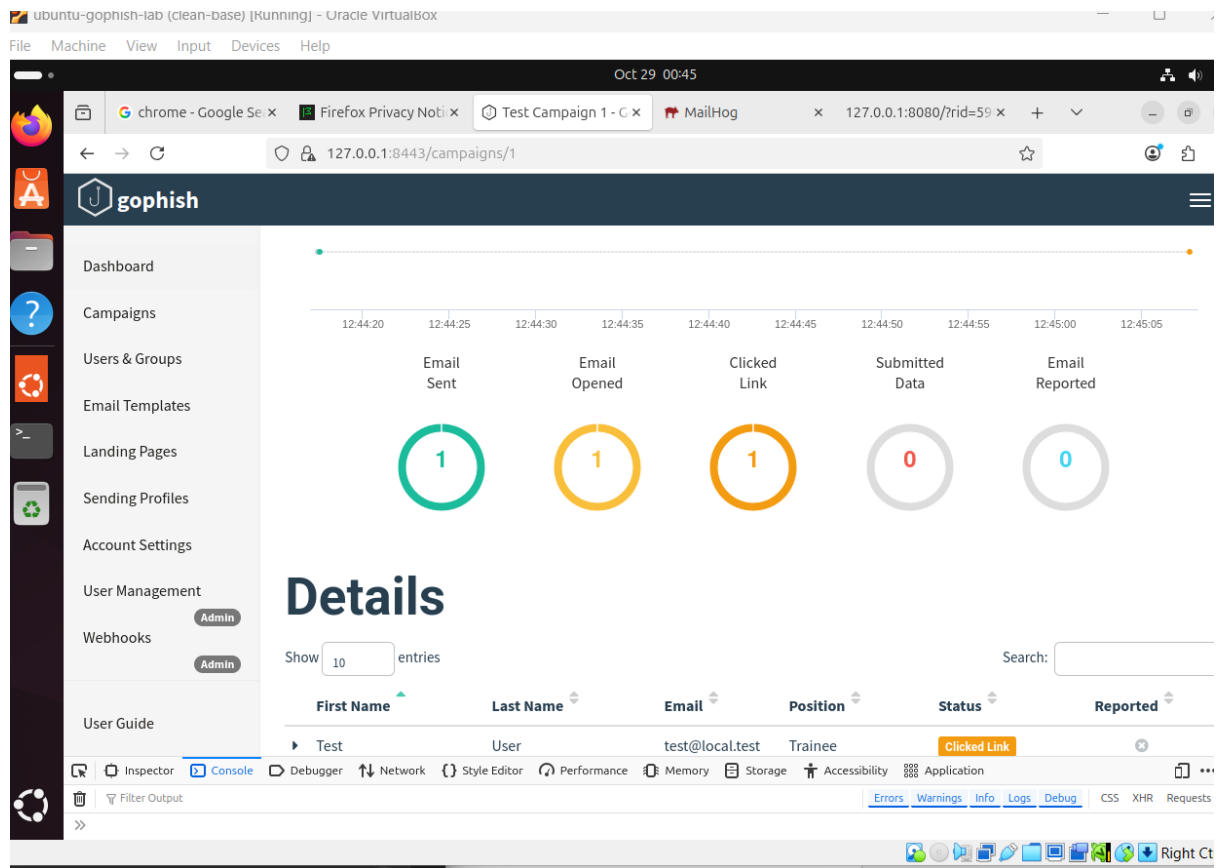
Showing 1 to 1 of 1 entries

[Previous](#)
[1](#)
[Next](#)

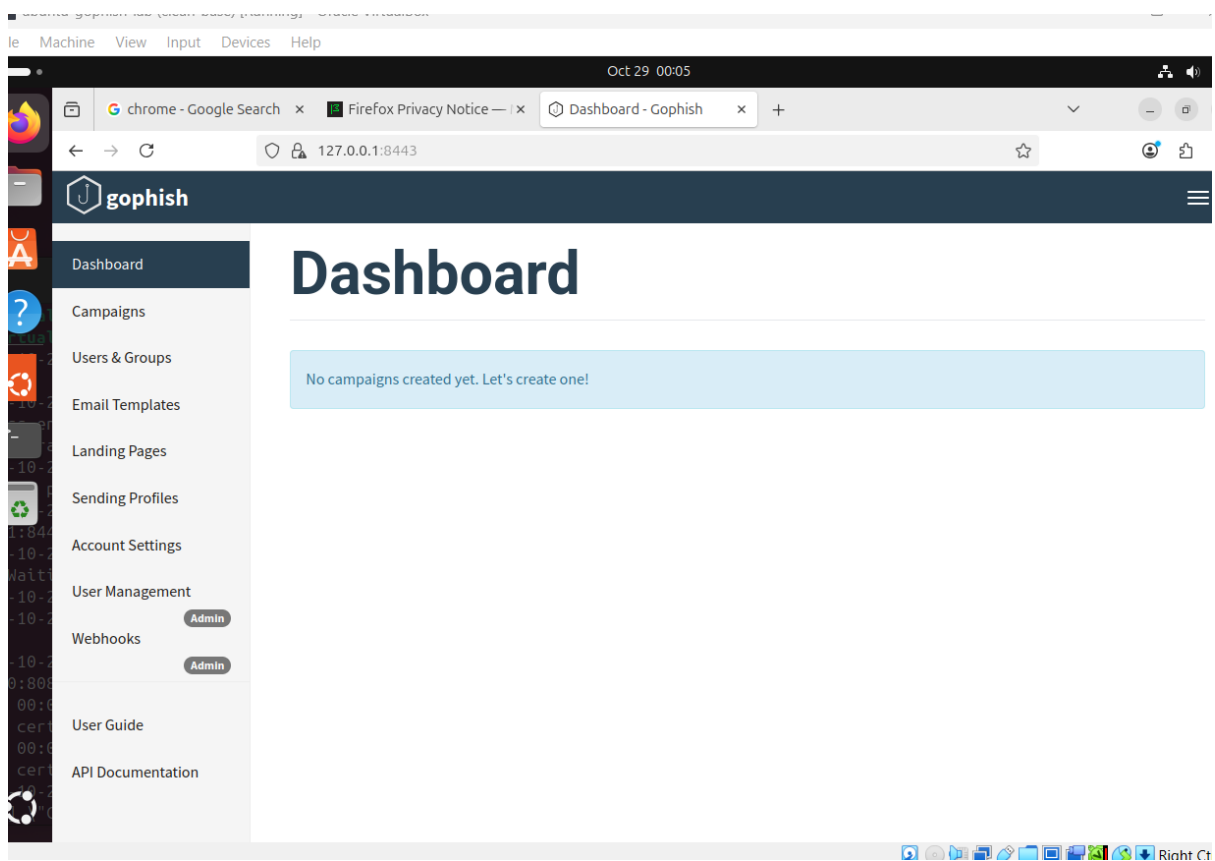
- Campaign:** Built and launched Lab_Campaign_1 using the template, landing page, sending profile, and test group.

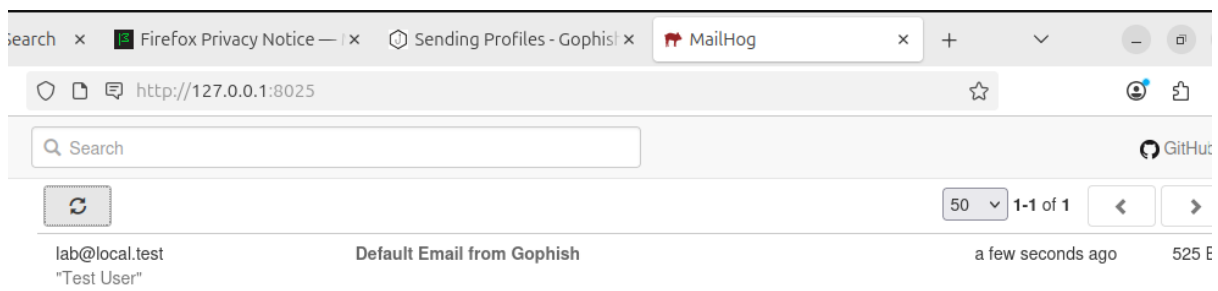
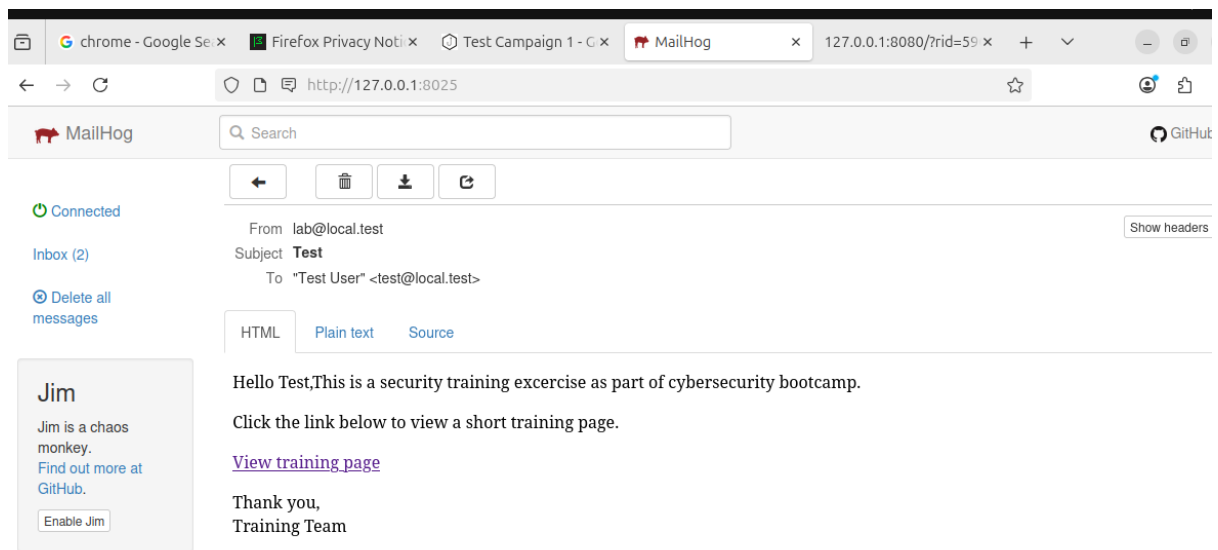
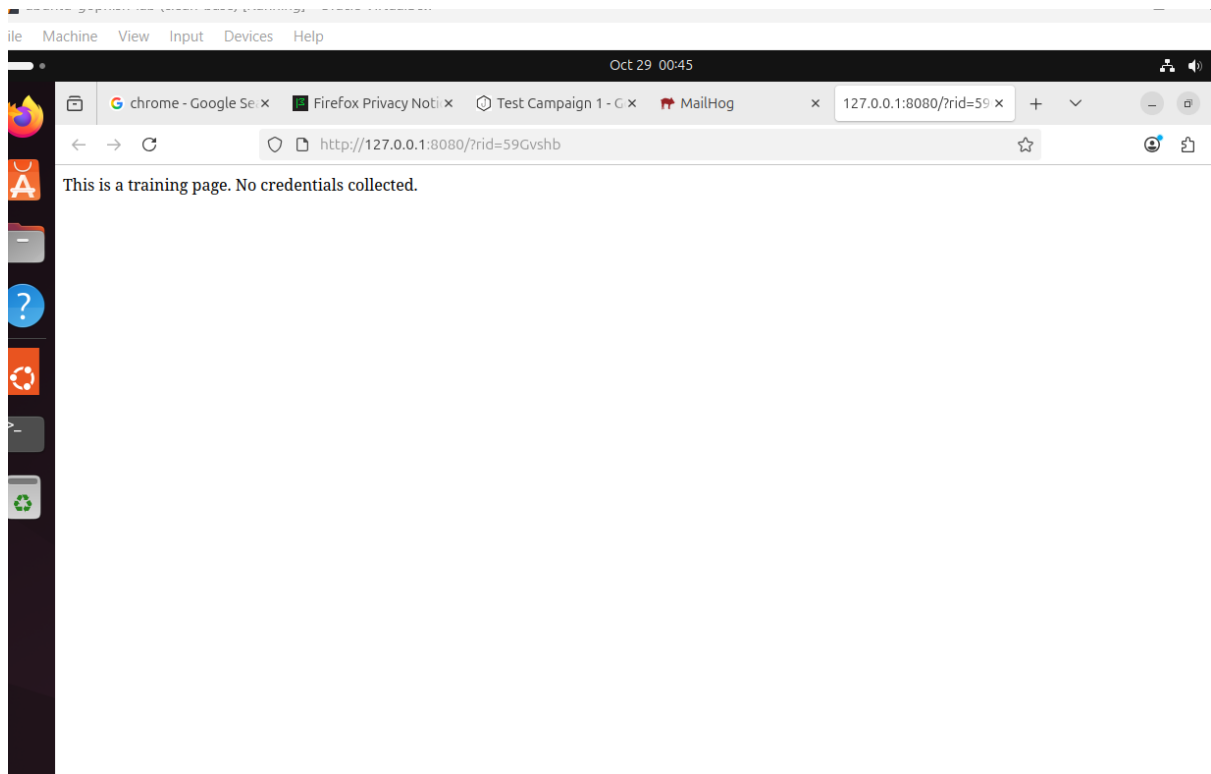


- Verification:** Monitored MailHog and GoPhish dashboard; validated Sent = 1, opened and clicked events when interacting with the captured message.



11. **Documentation:** Captured screenshots of each major step for reporting and reverted the VM to clean-base when finished.





5. Results (Metrics)

Metric	Count
Emails sent	1
Delivered (captured locally)	1
Opened	1
Clicked	1
Credentials submitted	0

6. Findings & Analysis

- The GoPhish → MailHog setup demonstrated the typical phishing workflow: message composition, sending, delivery capture, user interaction, and event logging.
- MailHog captured the message locally, preventing any external delivery — confirming safety controls worked.
- The test user clicked the landing page link; because credential capture was disabled for safety, no sensitive data was recorded.
- The exercise shows how easily a realistic-looking email can be delivered within an environment and how tracking and landing pages can be used for training.

7. Recommendations

Immediate (0–7 days)

- Provide targeted awareness training to test participants explaining the simulated campaign and lessons learned.
- Ensure a reporting mechanism exists (e.g., a “Report Phish” button) and that staff know how to use it.

Short term (1–3 months)

- Review and enable email authentication in production (SPF, DKIM, DMARC).
- Implement automated phishing reporting and triage workflows.

Long term (>3 months)

- Schedule recurring simulated campaigns (with explicit authorization) and measure improvement over time.
- Integrate user-training modules and track completion and improvement metrics.

